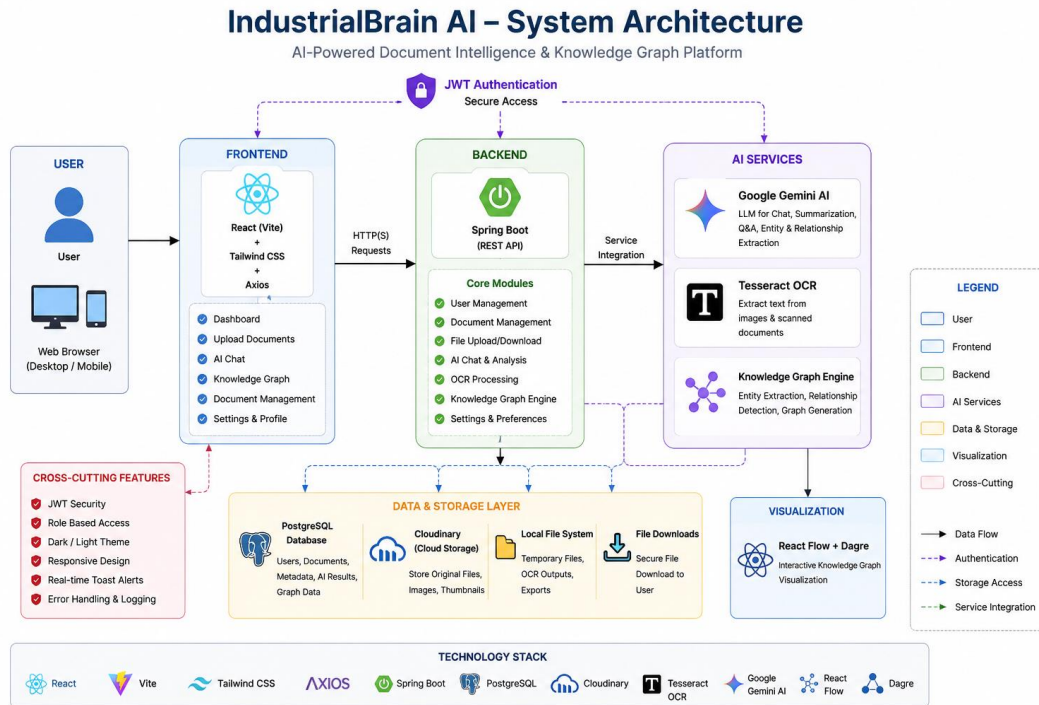


IndustrialBrain AI

AI-Powered Industrial Knowledge Intelligence Platform



Economic Times AI Hackathon 2.0

Project Documentation

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Submission Year

2026

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1. Executive Summary

IndustrialBrain AI is an AI-powered industrial knowledge intelligence platform designed to simplify the management and understanding of industrial documents. Organizations often store critical information in PDFs, scanned reports, manuals, and images, making knowledge retrieval slow and inefficient.

The platform combines Optical Character Recognition (OCR), Artificial Intelligence, and Knowledge Graph technology into one solution. Users can securely upload industrial documents, extract text from scanned files, interact with documents using Google Gemini AI, and visualize relationships through an interactive Knowledge Graph.

Built using React, Spring Boot, PostgreSQL, Google Gemini AI, and Tesseract OCR, the platform provides secure document management, intelligent search, AI-powered explanations, and visual knowledge discovery. Its modular architecture ensures scalability, maintainability, and future extensibility.

2. Project Highlights

- AI-powered document understanding
- OCR for scanned documents
- Interactive knowledge graphs
- JWT authentication
- Cloudinary integration
- React + Spring Boot architecture
- PostgreSQL database

3. Problem Statement

Industrial organizations generate thousands of technical manuals, maintenance reports, standard operating procedures, compliance documents, inspection reports, and engineering drawings. Most of this information remains isolated inside PDFs and scanned documents, making it difficult for employees to quickly locate critical knowledge.

Traditional document management systems primarily provide storage without intelligent understanding. Valuable information becomes fragmented, manual searches consume significant time, and important relationships between entities remain hidden.

IndustrialBrain AI transforms static industrial documents into an intelligent knowledge ecosystem using OCR, Generative AI, and Knowledge Graph visualization. The platform enables users to extract structured information, ask contextual questions, explore relationships, and improve decision-making.

4. Proposed Solution

IndustrialBrain AI provides a centralized platform for intelligent industrial document management by integrating OCR, Generative AI, and Knowledge Graph technologies. The platform converts unstructured documents into searchable digital knowledge, enabling engineers and professionals to retrieve information quickly and accurately.

Users can upload PDF files and scanned images, extract text using Tesseract OCR, interact with documents through Google Gemini AI, and explore relationships between entities using an interactive Knowledge Graph. The system is designed with a modular architecture to ensure scalability, maintainability, and seamless integration with enterprise environments.

Objectives

Digitize industrial documents using OCR.

Provide AI-powered document understanding.

Generate interactive knowledge graphs.

Enable secure document storage with authentication.

Improve productivity through intelligent search and visualization.

5. Key Features

Secure Authentication

JWT-based login, registration, and protected routes.

Document Management

Upload, preview, download, and delete industrial documents.

OCR Processing

Extract text from scanned PDFs and images using Tesseract OCR.

AI Assistant

Ask questions and receive contextual answers powered by Google Gemini AI.

Knowledge Graph

Automatically visualize relationships between extracted entities.

Interactive Graph

Search, highlight nodes, reset view, and export the graph as PNG.

Dashboard Analytics

View document statistics, upload trends, and recent activity.

Theme Support

Light and Dark mode for improved accessibility.

Cloud Storage

Store uploaded files securely using Cloudinary.

Responsive UI

Optimized for desktop and modern web browsers.

6. Technology Stack

Layer	Technology
Frontend	React (Vite), Tailwind CSS, React Router, React Flow, Axios
Backend	Spring Boot, Spring Security, JWT Authentication, REST APIs
Database	PostgreSQL
Artificial Intelligence	Google Gemini AI, Tesseract OCR
Cloud Services	Cloudinary
Development Tools	Git, GitHub, Postman, VS Code, Maven
Architecture Pattern	Client-Server Architecture with RESTful APIs and Modular Design

7. System Architecture

IndustrialBrain AI follows a modular client-server architecture. The React frontend communicates with Spring Boot REST APIs using JWT-secured requests. The backend coordinates document storage, OCR processing, AI interactions, knowledge graph generation, and PostgreSQL persistence. Cloudinary stores uploaded documents, while Google Gemini AI provides intelligent responses.

Architecture Diagram

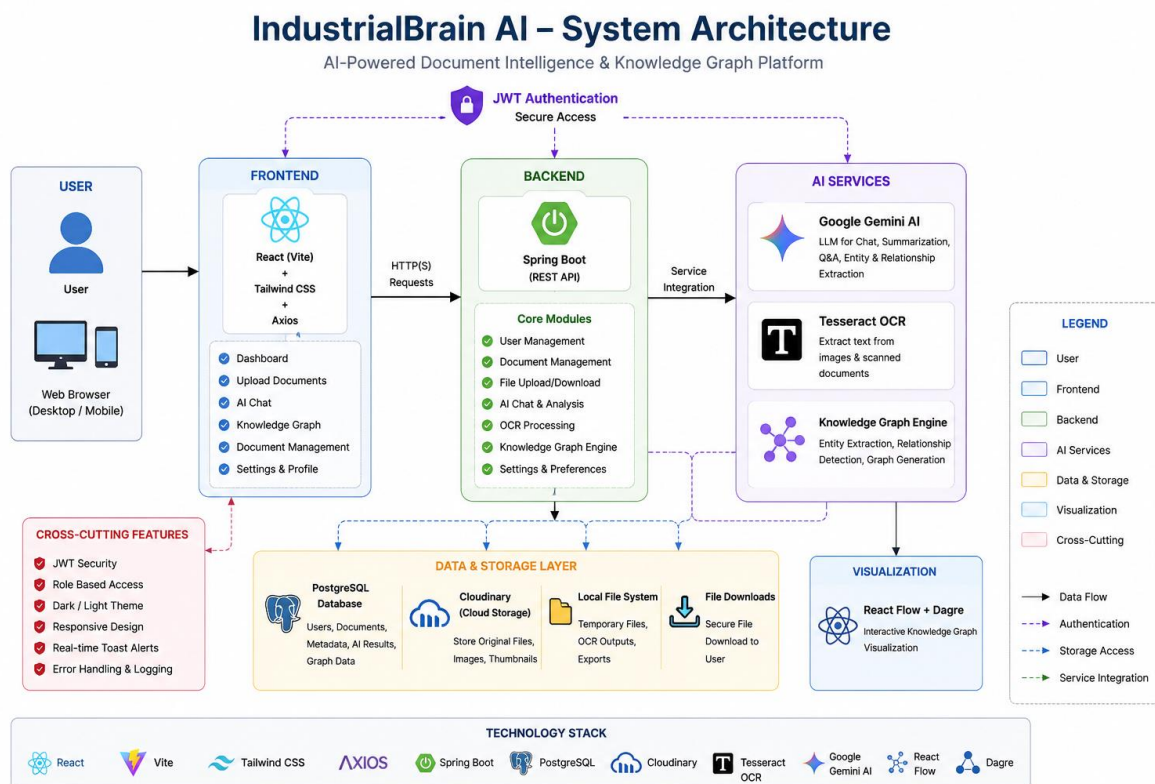


Figure 1: Overall System Architecture showing interactions between the client, backend services, AI modules, cloud storage, and database.

8. Application Workflow

The application workflow begins with user authentication, followed by secure document upload. Uploaded files are processed through OCR to extract text. The extracted content is stored in PostgreSQL and becomes available for AI-powered querying and knowledge graph generation. Users can manage documents, visualize entity relationships, and download results.

Workflow Diagram

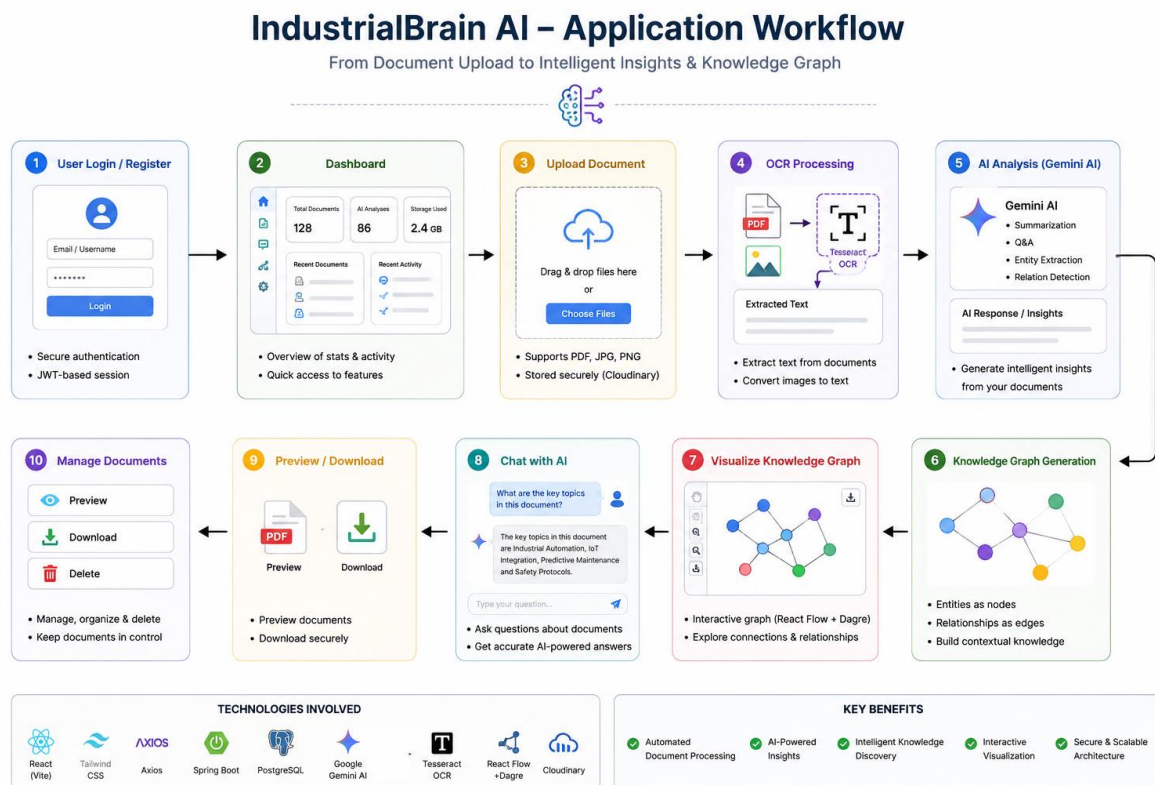


Figure 2: End-to-end workflow from user login to AI analysis and visualization.

9. AI Processing Pipeline

The AI pipeline integrates OCR and Google Gemini AI to transform raw industrial documents into structured knowledge. OCR extracts machine-readable text from scanned files. The extracted content is then analyzed by Gemini AI to generate summaries, answer user queries, identify entities, and build relationships for knowledge graph visualization.

Pipeline Diagram

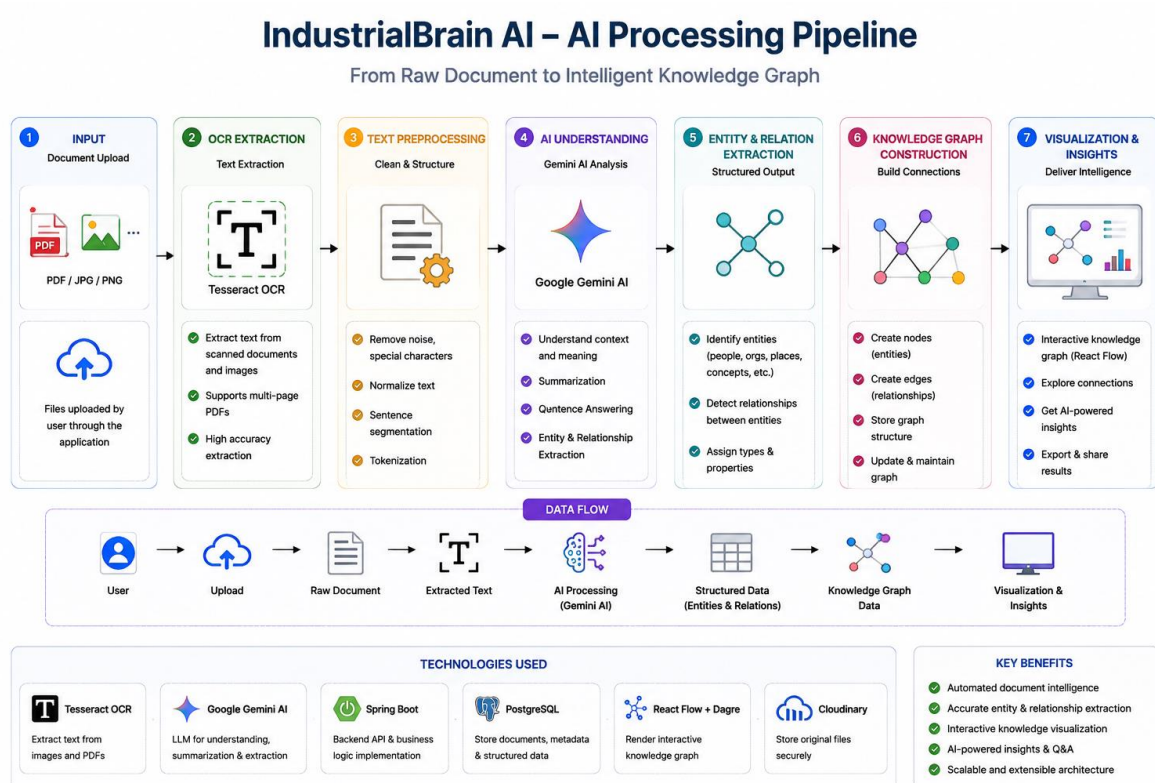


Figure 3: OCR, AI analysis, entity extraction, and knowledge graph generation pipeline.

10. Database Design

The backend uses PostgreSQL to manage users, uploaded documents, extracted OCR text, AI-generated responses, and knowledge graph metadata. A normalized relational schema ensures data consistency, security, and efficient querying.

ER Diagram

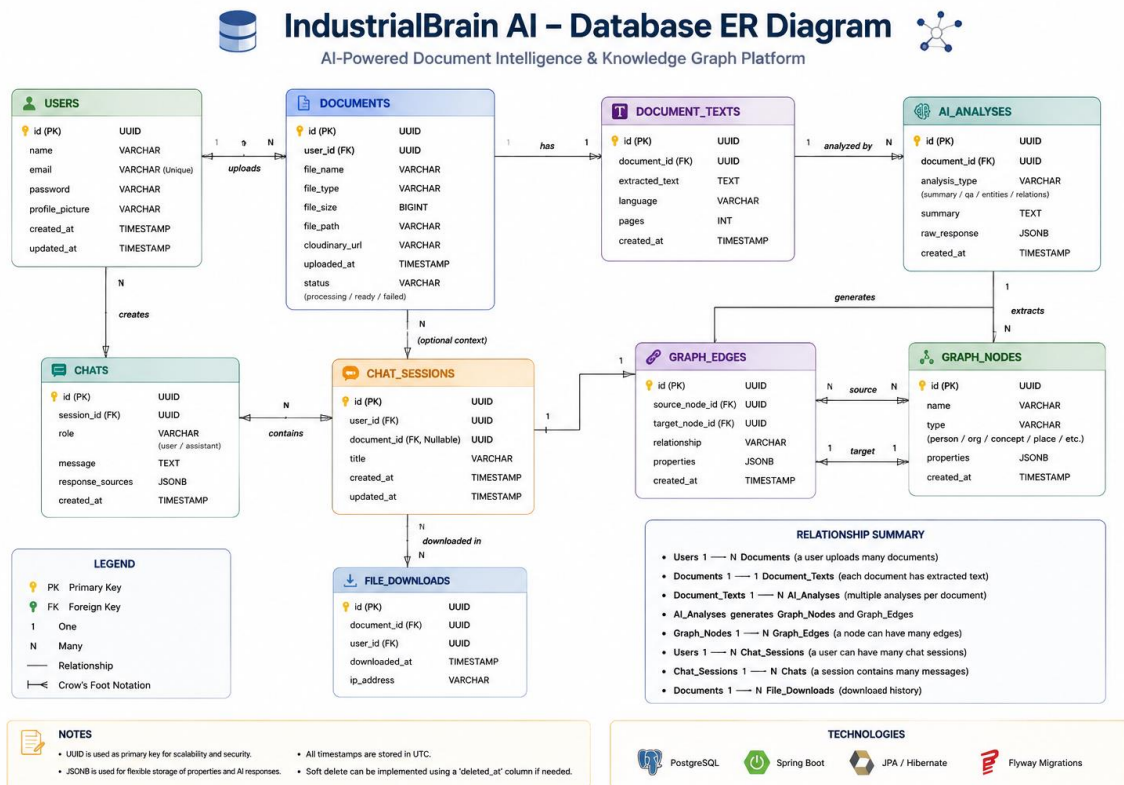


Figure 4: Entity Relationship Diagram illustrating the database schema used by IndustrialBrain AI.

11. Implementation Details

Frontend

The frontend is developed using React (Vite) and Tailwind CSS. It provides a responsive user interface with secure authentication, dashboard analytics, document management, AI chat, and interactive knowledge graph visualization.

Backend

Spring Boot powers the REST APIs, authentication, OCR integration, AI services, and business logic. JWT-based security protects all authenticated endpoints.

Database

PostgreSQL stores user information, document metadata, OCR results, AI responses, and knowledge graph data in a structured relational database.

Cloud Storage

Cloudinary securely stores uploaded industrial documents and enables reliable retrieval for preview and download.

AI Integration

Google Gemini AI provides contextual document explanations, question answering, and intelligent analysis. Tesseract OCR extracts text from scanned PDFs and images.

12. Project Screenshots

The following screenshots demonstrate the key functionality implemented in IndustrialBrain AI.

Authentication

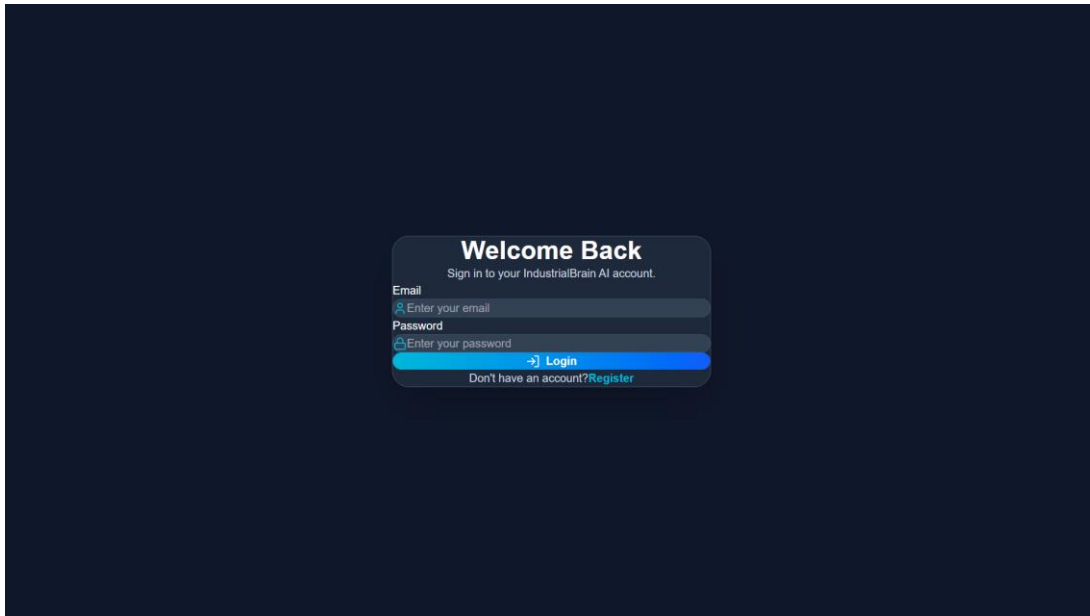


Figure 5: User Login Page

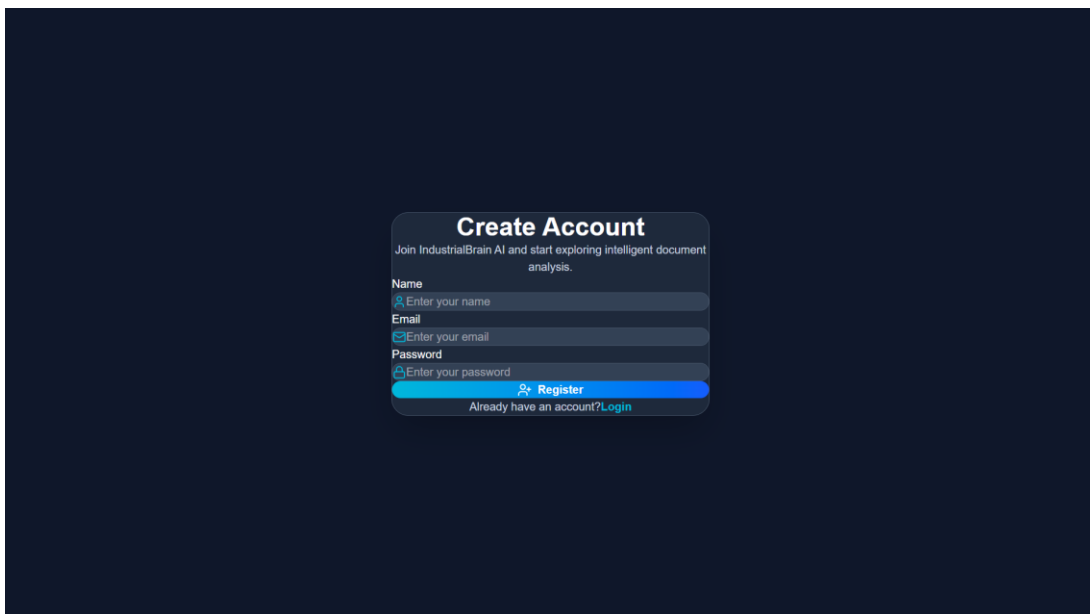


Figure 6: User Registration Page

Dashboard

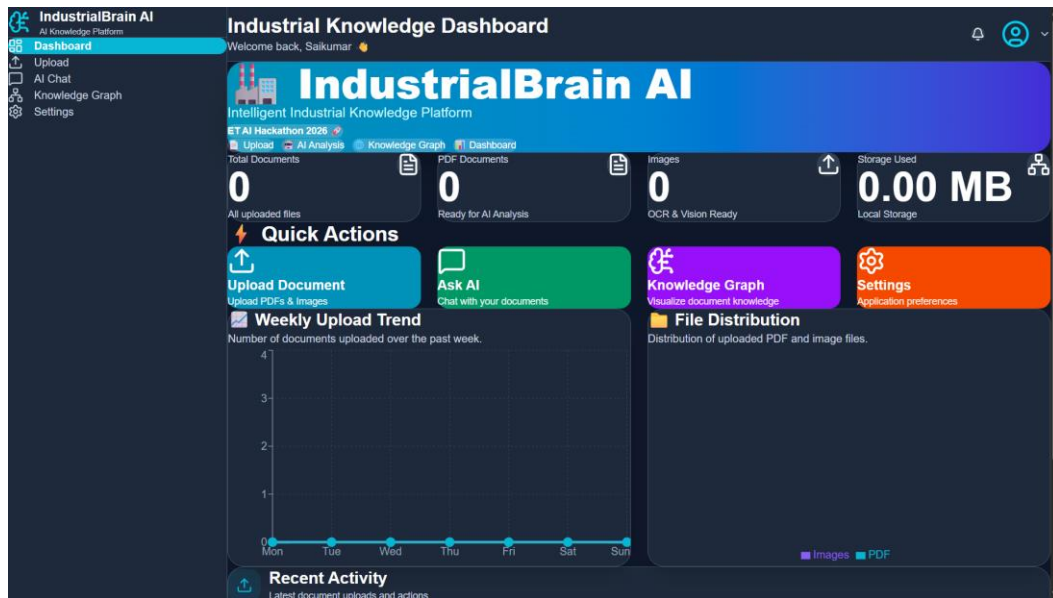


Figure 7: Dashboard Overview

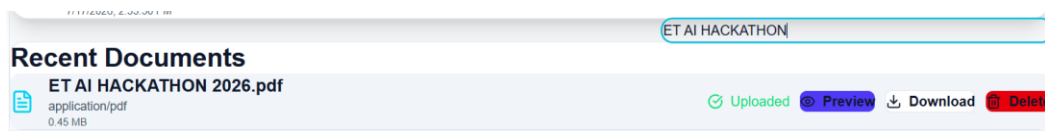


Figure 8: Dashboard Search Results

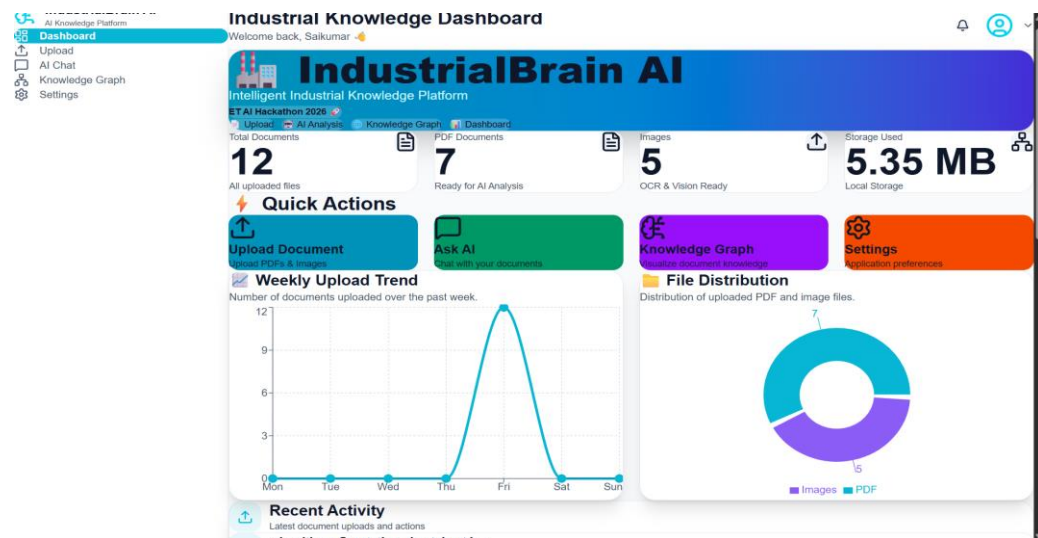


Figure 9: Dashboard Statistics

Document Management

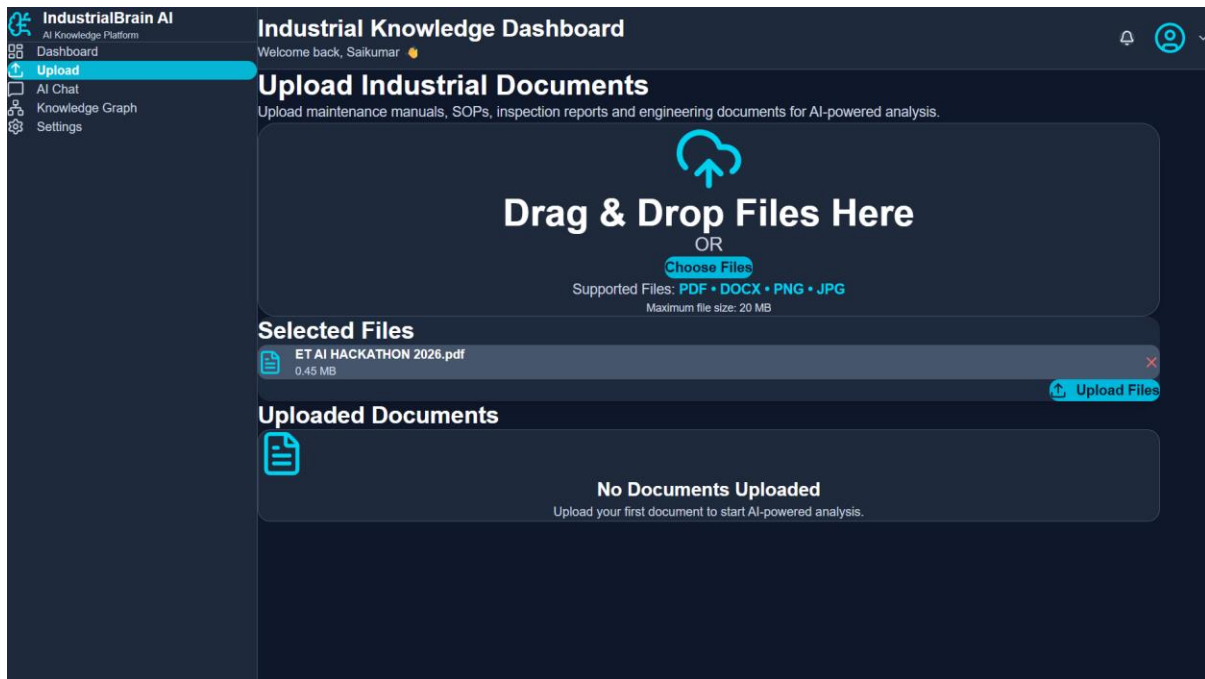


Figure 10: Upload Documents Interface

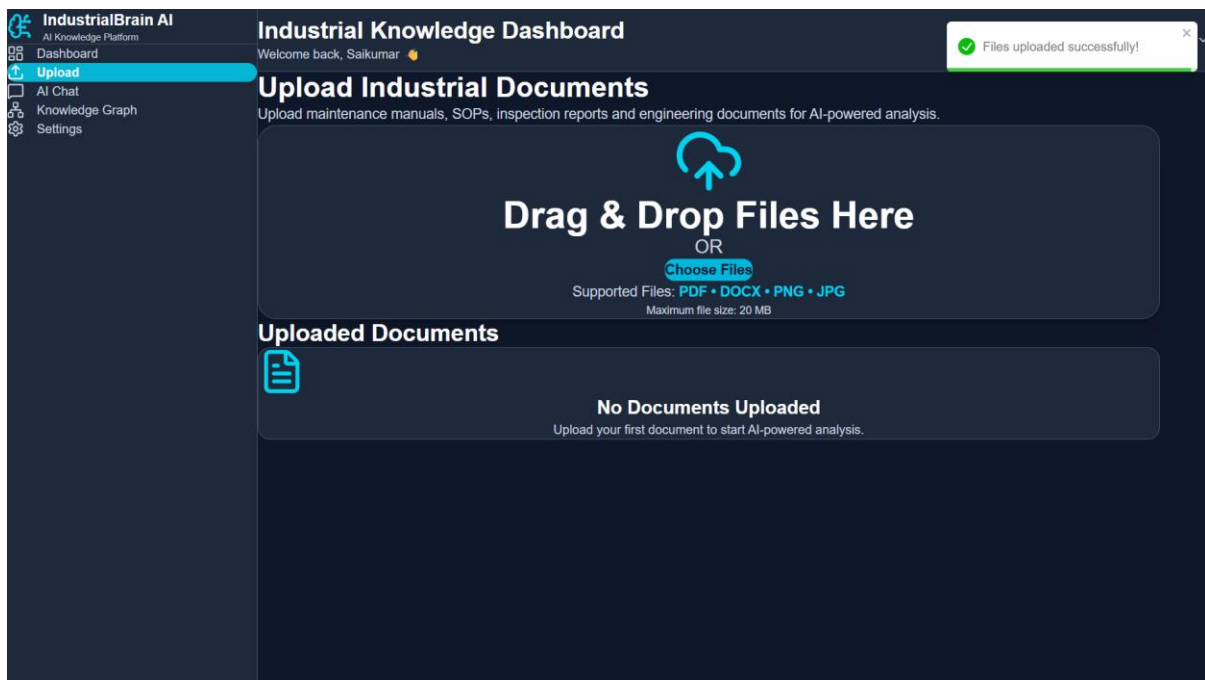


Figure 11: Document Upload Successful

Uploaded Documents			
	ET AI HACKATHON 2026.pdf	0.45 MB	   
	OCR_Scanned_Document.pdf	0.82 MB	   
	OCR_Scan_Image.png	0.10 MB	   
	100 DSA Interview Questions you can't afford to miss-.pdf	0.05 MB	   
	GAP YEAR OPPORTUNITIES.pdf	0.19 MB	   
	OOPS Interview Questions.pdf	3.11 MB	   
	LOGIC BUILDING WITH LOOPS.pdf	0.33 MB	   
	DSA_ROADMAP.pdf	0.01 MB	   
	Java-collections-cheat-sheet4.jpg	0.08 MB	   
	dfs-and-bfs-cheatsheet1.jpg	0.09 MB	   
	recursion-cheatsheet1.jpg	0.04 MB	   
	sorting-algorithms-O-notation-cheatsheet.jpg	0.03 MB	   
	algorithms-O-notation-cheatsheet.jpg	0.08 MB	   
	data-structures-O-notation-cheatsheet.jpg	0.06 MB	   

Figure 12: Document List


IndustrialBrain AI
AI Knowledge Platform

Dashboard
Upload
AI Chat
Knowledge Graph
Settings

Industrial Knowledge Dashboard

Welcome back, Saikumar 🌟

Document Preview

Previous Page 1 / 17 Next

ET AI HACKATHON 2026

Problem Statements

Industrial | Smart Cities | Sustainability | Supply Chain | Geospatial | Public Safety | Sales Excellence

Each team must select one problem statement. Choose the challenge that best aligns with your team's skills, expertise, and interests.

1
AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

Theme: Industrial Intelligence / Worker Safety / Geospatial Safety Analytics

PROBLEM CONTEXT

India's heavy industrial sector continues to pay a devastating human cost. According to DGFASLI, over 6,500 fatal workplace accidents were recorded in FY2023 — and that figure excludes most of the mining and construction sectors. In one of the most disturbing recent incidents, eight workers died at Visakhapatnam Steel Plant in January 2025 when entrapped gases triggered a sudden explosion in the coke oven battery — a facility that had functioning safety systems including gas detectors, permit-to-work controls, and SCADA. An investigation by The Wire found that warning signals from gas pressure sensors existed, but no intelligence layer connected those readings to operational decisions in time. This pattern — data present, but unacted upon — repeats itself across Indian heavy industry. A FICCI

Figure 13: Document Preview Page

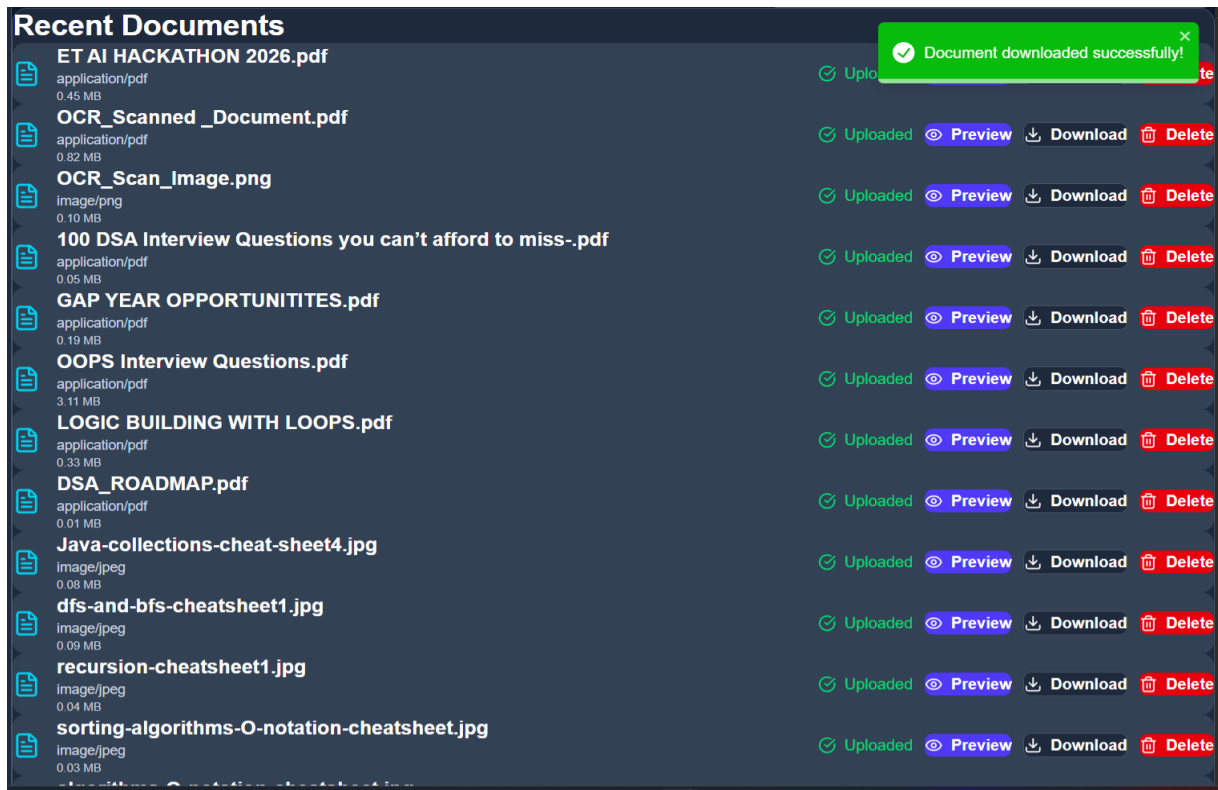


Figure 14: Document Download

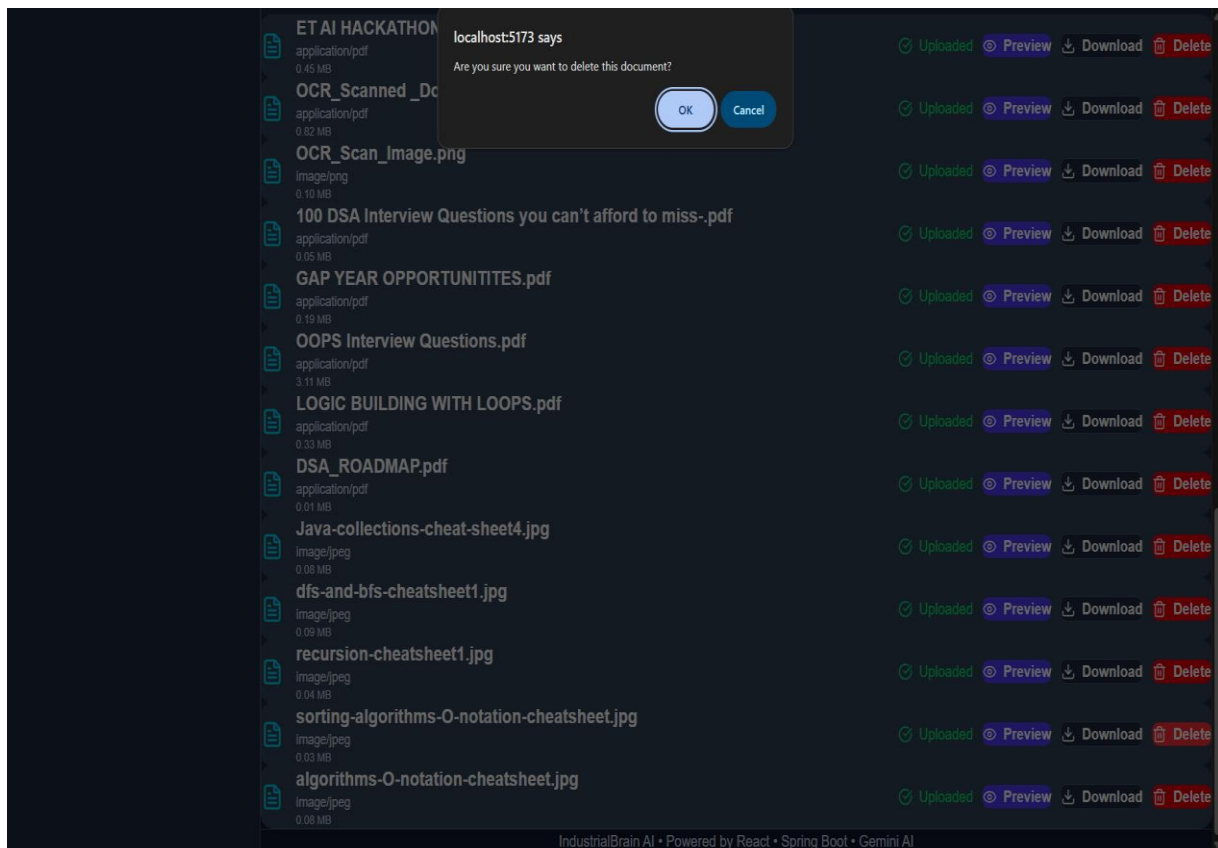


Figure 15: Document Delete Dialog Result

Artificial Intelligence

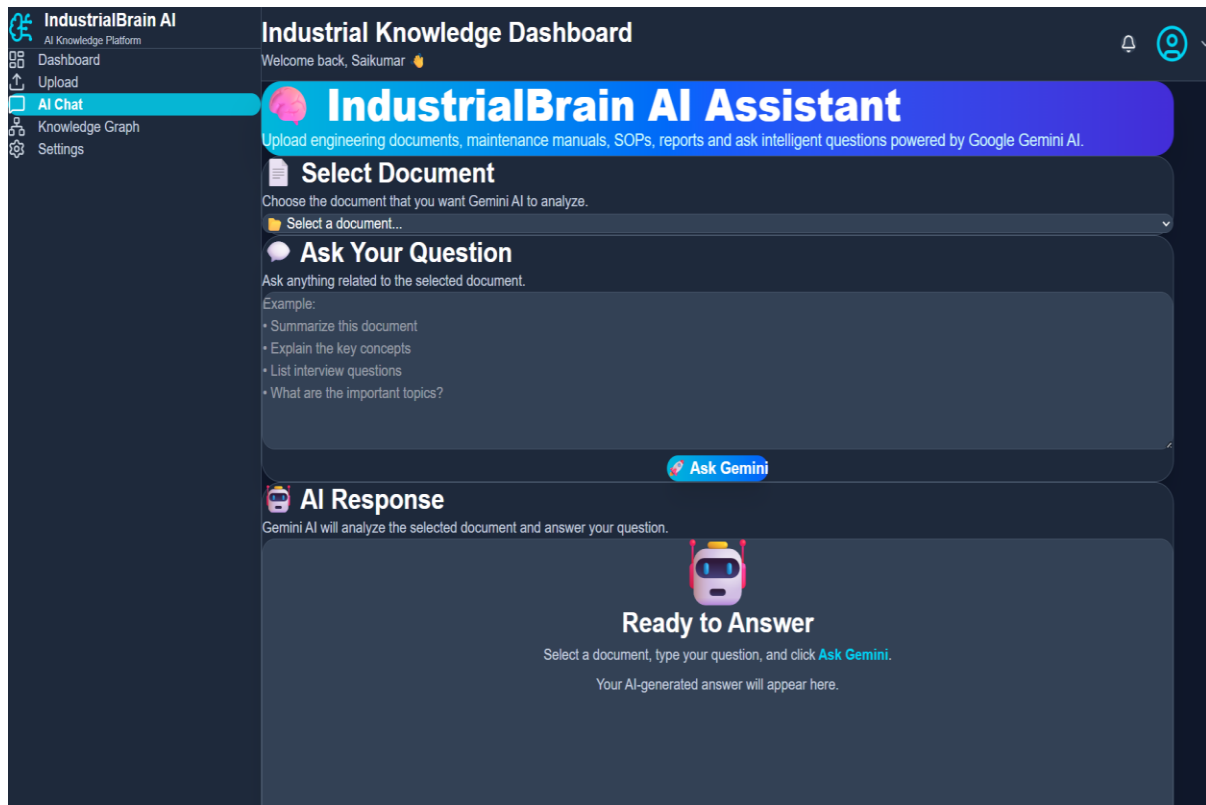


Figure 16: AI Chat Interface

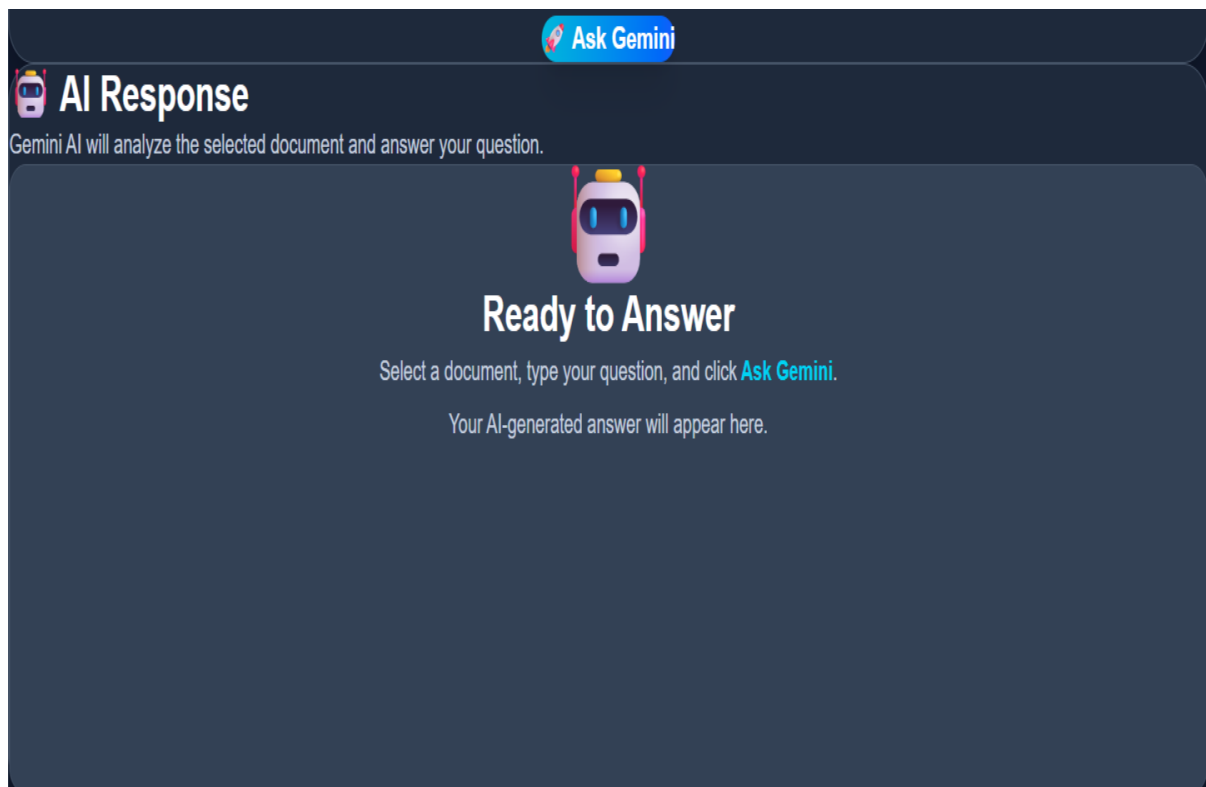


Figure 17: AI Generated Response

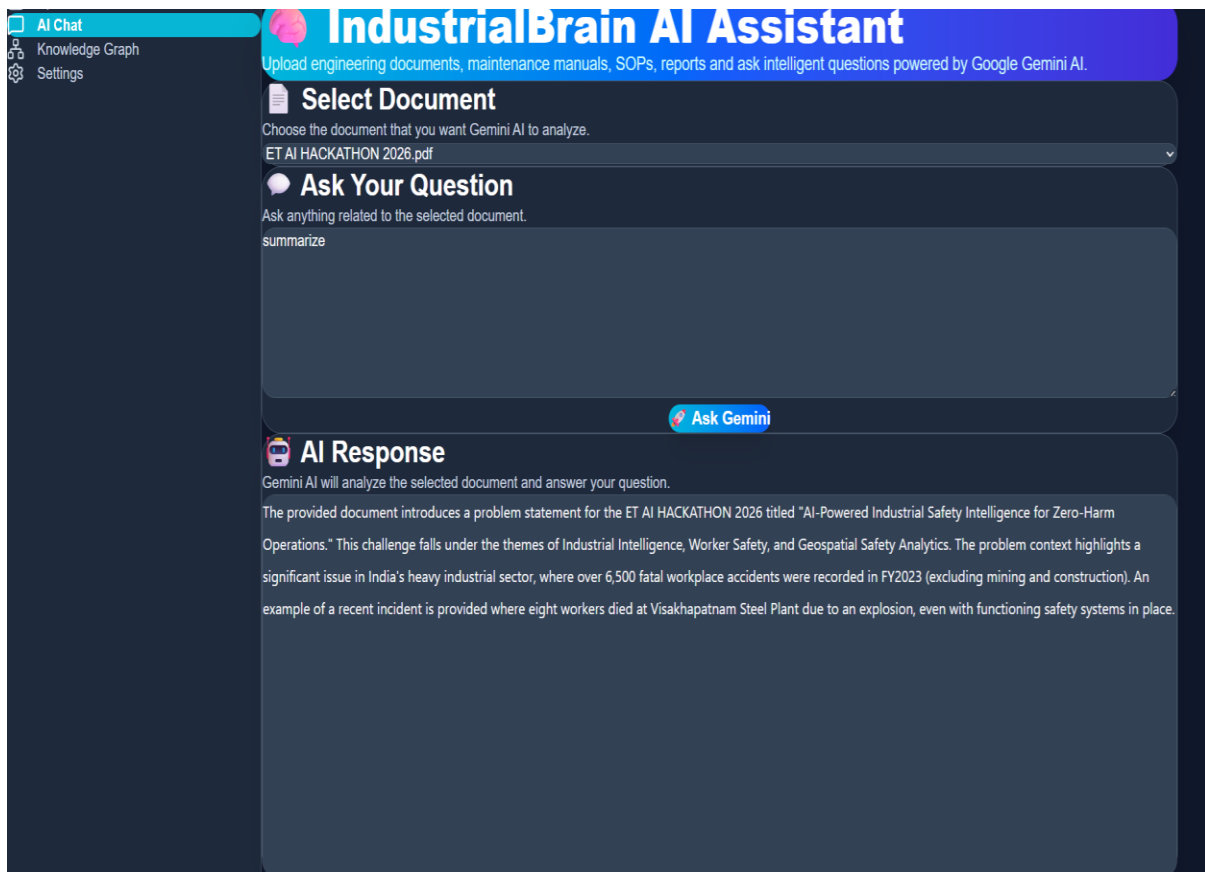


Figure 18: AI Document Explanation

Knowledge Graph

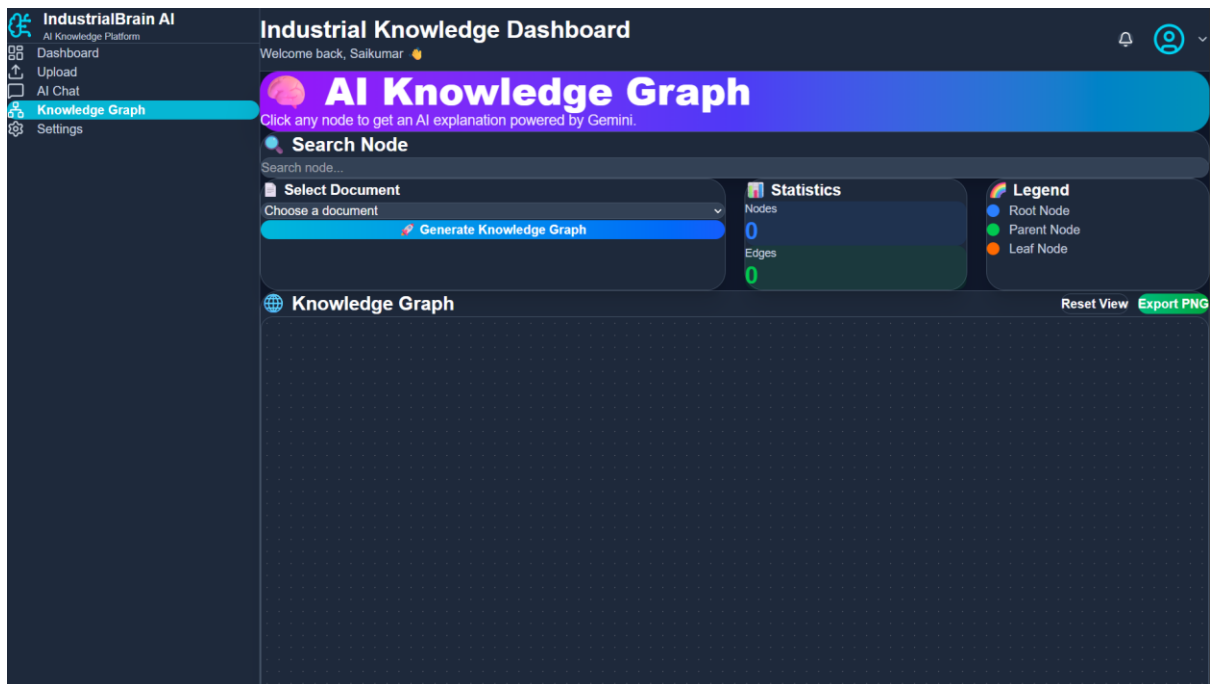


Figure 19: Knowledge Graph Overview

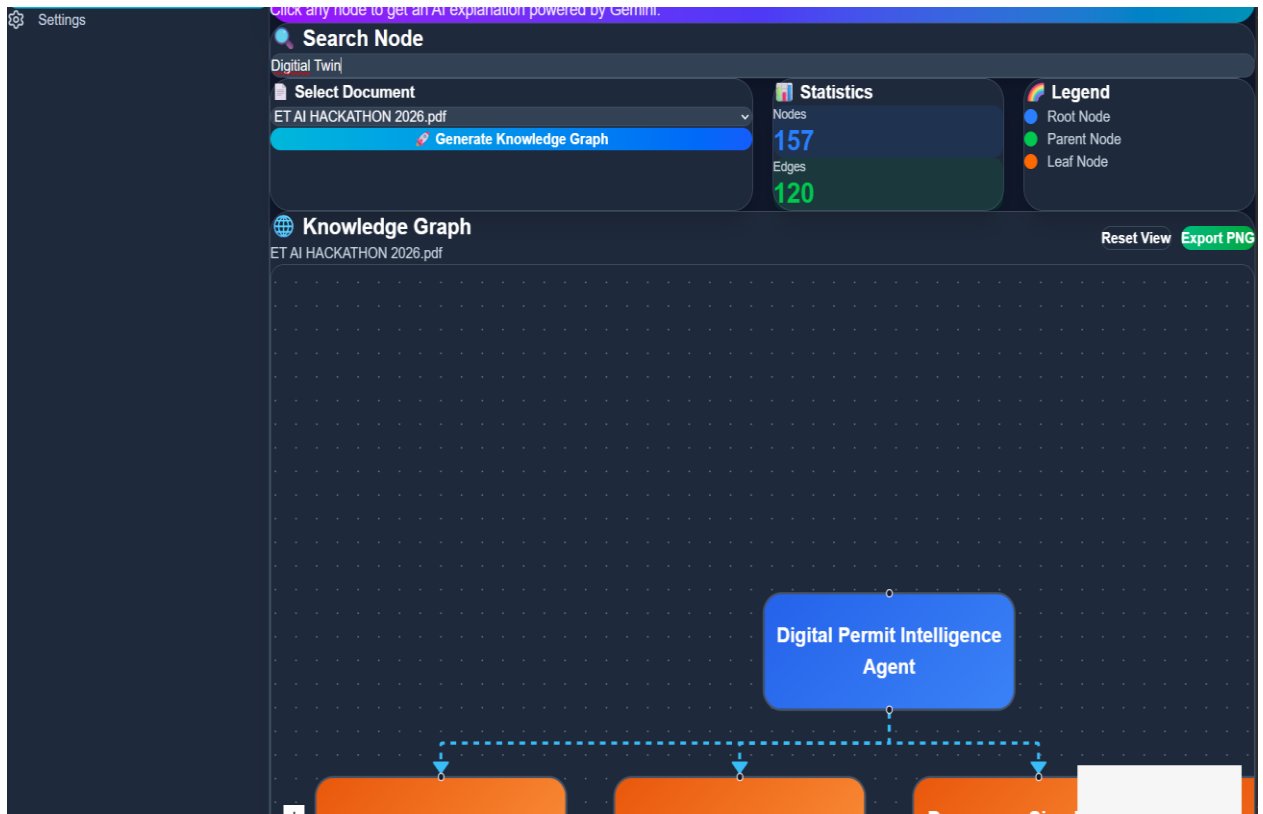


Figure 20: Knowledge Graph Search

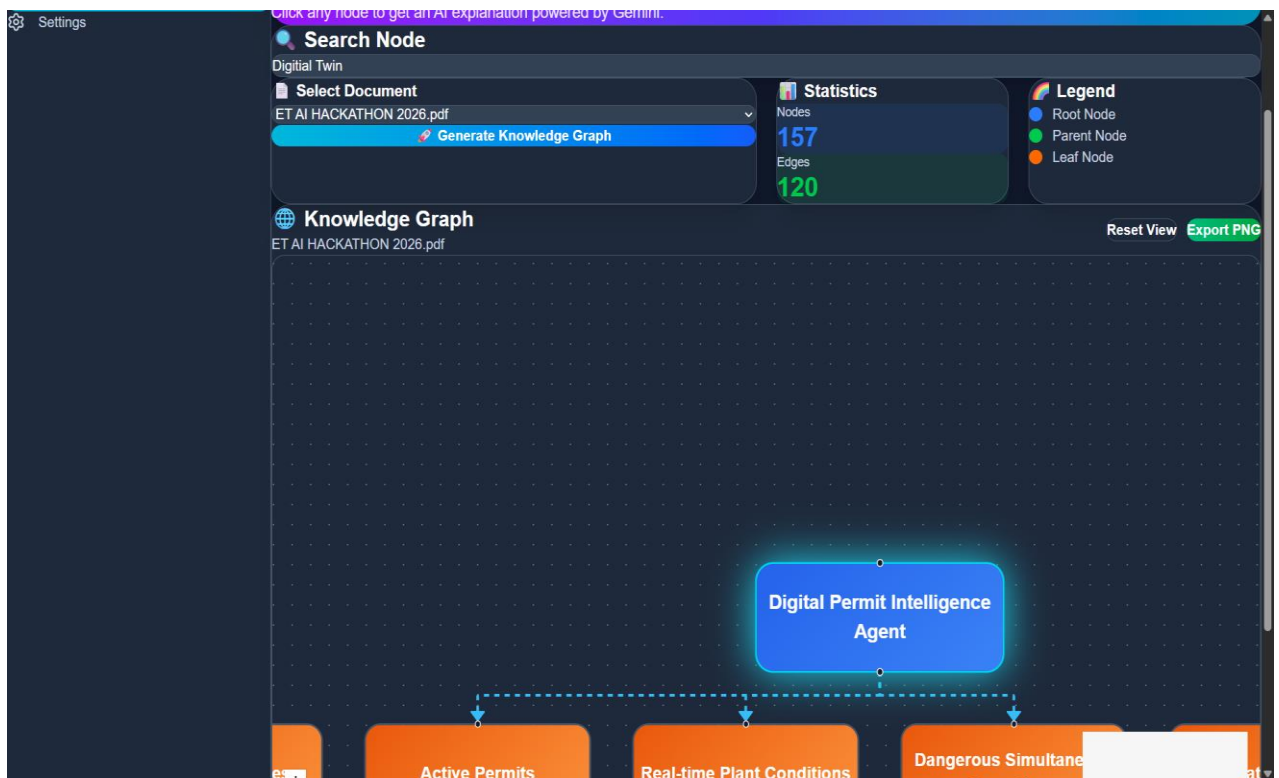
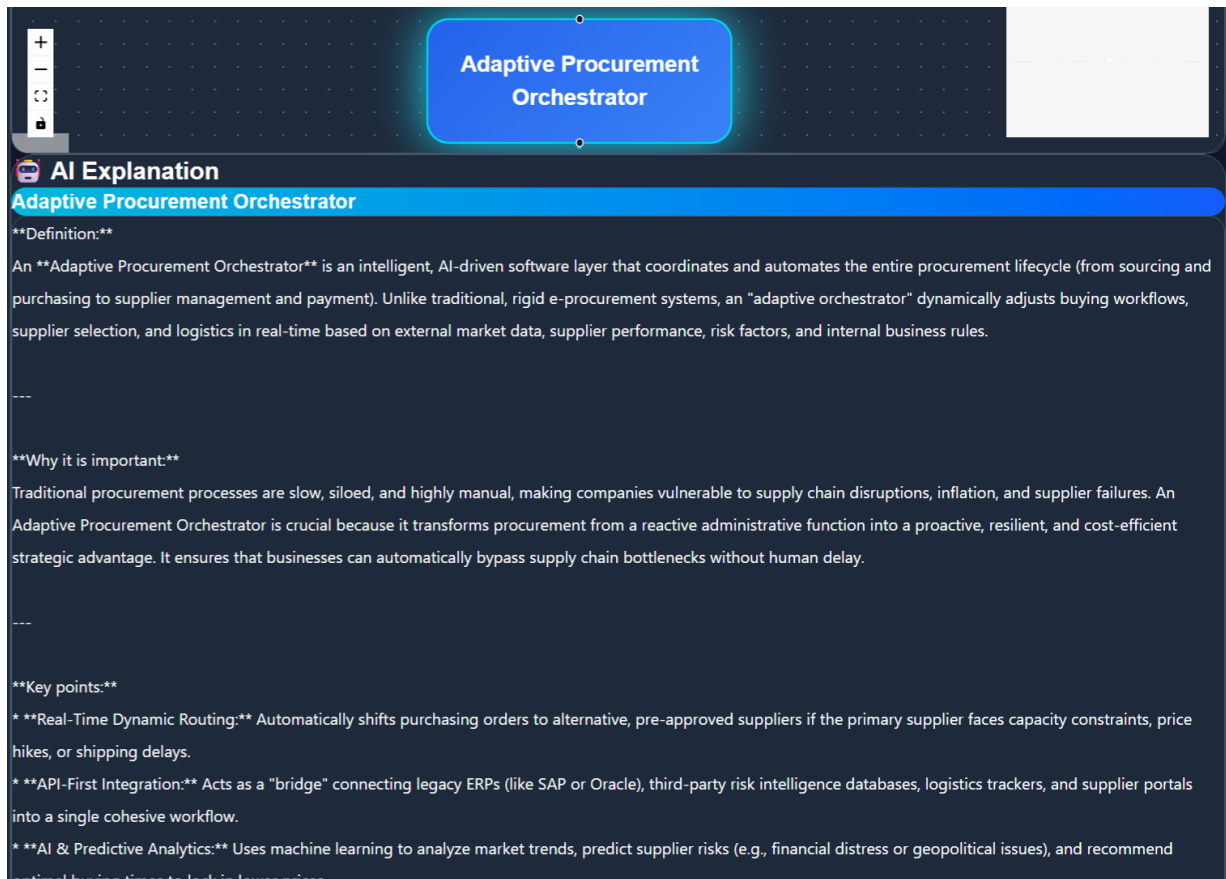


Figure 21: Knowledge Graph Node Highlight



Adaptive Procurement Orchestrator

AI Explanation

Adaptive Procurement Orchestrator

****Definition:****

An **Adaptive Procurement Orchestrator** is an intelligent, AI-driven software layer that coordinates and automates the entire procurement lifecycle (from sourcing and purchasing to supplier management and payment). Unlike traditional, rigid e-procurement systems, an "adaptive orchestrator" dynamically adjusts buying workflows, supplier selection, and logistics in real-time based on external market data, supplier performance, risk factors, and internal business rules.

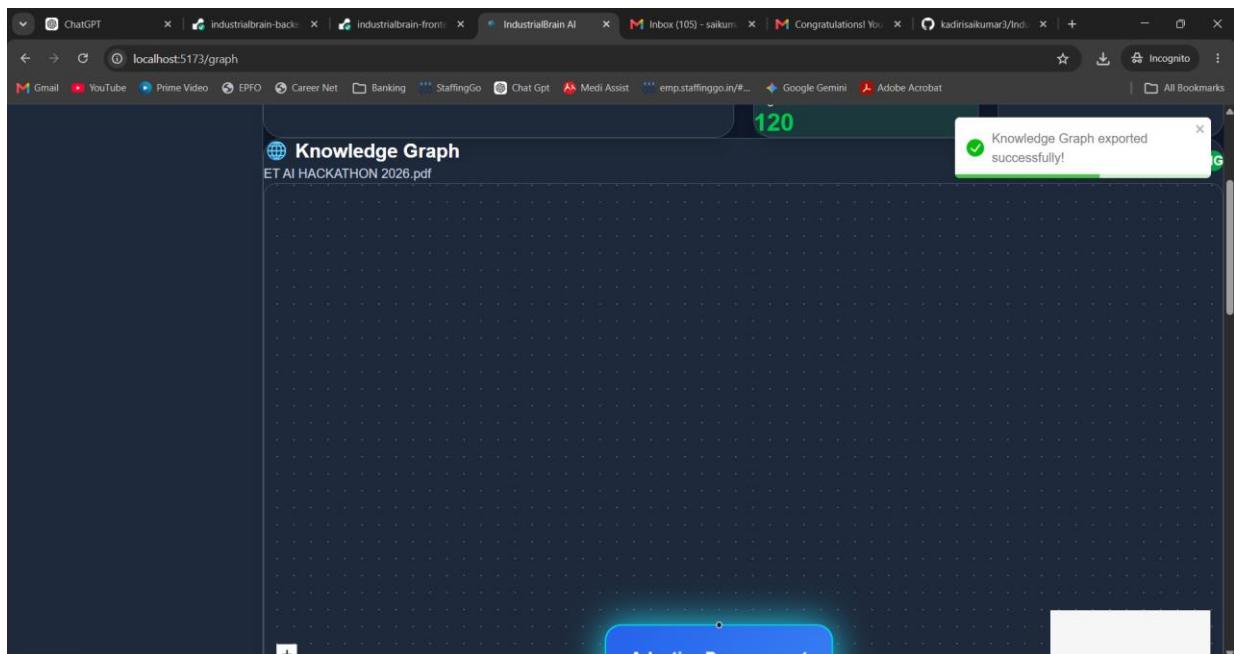
****Why it is important:****

Traditional procurement processes are slow, siloed, and highly manual, making companies vulnerable to supply chain disruptions, inflation, and supplier failures. An Adaptive Procurement Orchestrator is crucial because it transforms procurement from a reactive administrative function into a proactive, resilient, and cost-efficient strategic advantage. It ensures that businesses can automatically bypass supply chain bottlenecks without human delay.

****Key points:****

- **Real-Time Dynamic Routing:**** Automatically shifts purchasing orders to alternative, pre-approved suppliers if the primary supplier faces capacity constraints, price hikes, or shipping delays.
- **API-First Integration:**** Acts as a "bridge" connecting legacy ERPs (like SAP or Oracle), third-party risk intelligence databases, logistics trackers, and supplier portals into a single cohesive workflow.
- **AI & Predictive Analytics:**** Uses machine learning to analyze market trends, predict supplier risks (e.g., financial distress or geopolitical issues), and recommend optimal buying times to lock in lower prices.

Figure 22: AI Explanation for Selected Node



Knowledge Graph
ET AI HACKATHON 2026.pdf

120

Knowledge Graph exported successfully!

Adaptive Procurement

Figure 23: Knowledge Graph Export

Profile & Settings

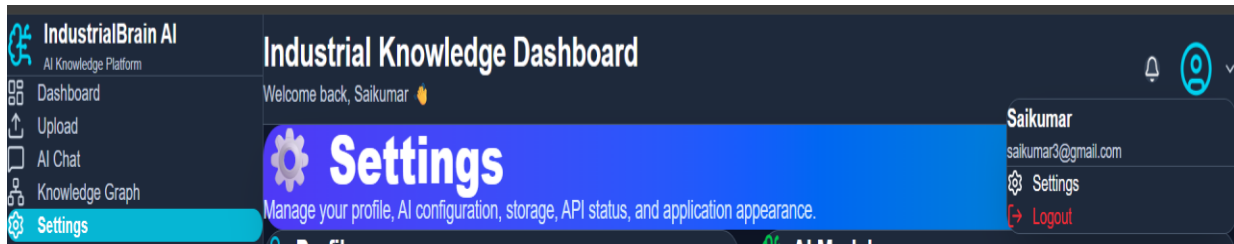


Figure 24: Profile Menu

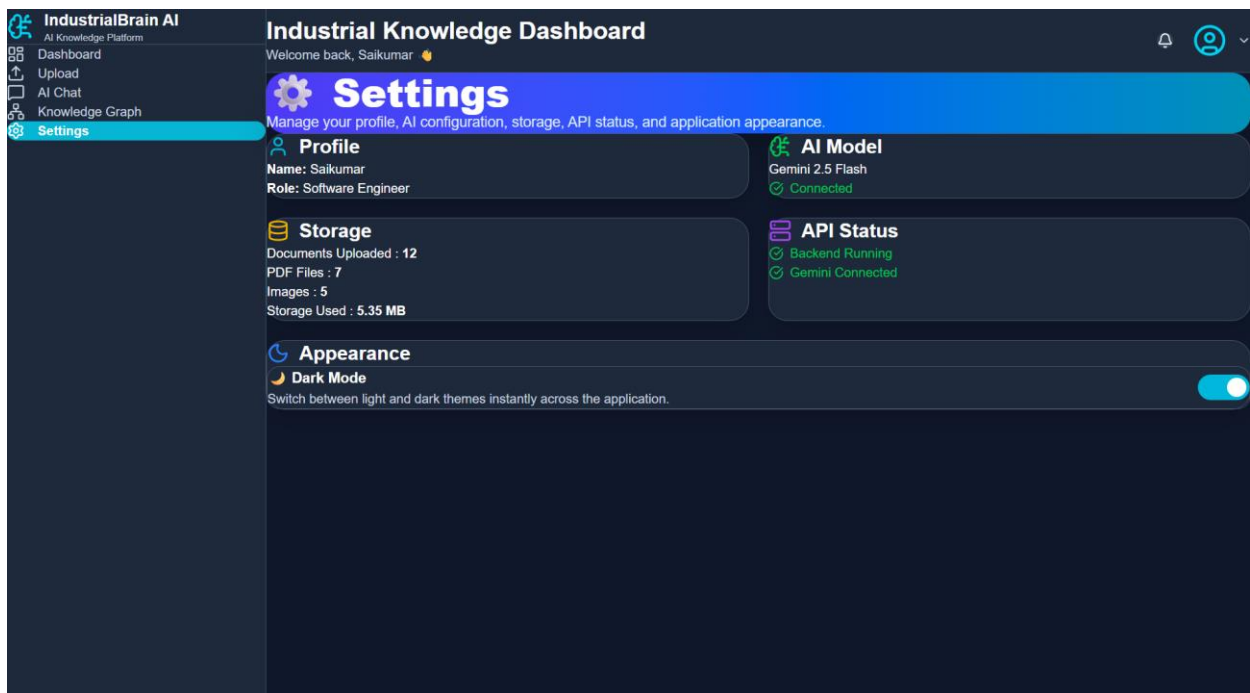


Figure 25: Settings Page

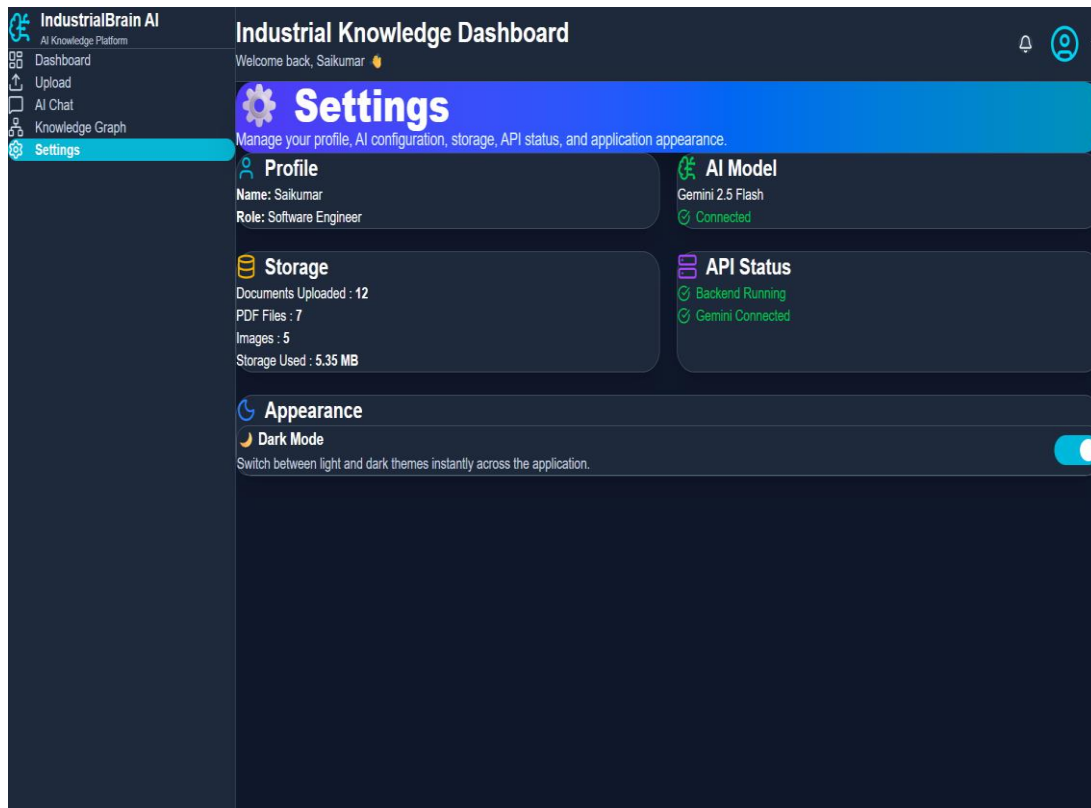


Figure 26: Dark Theme

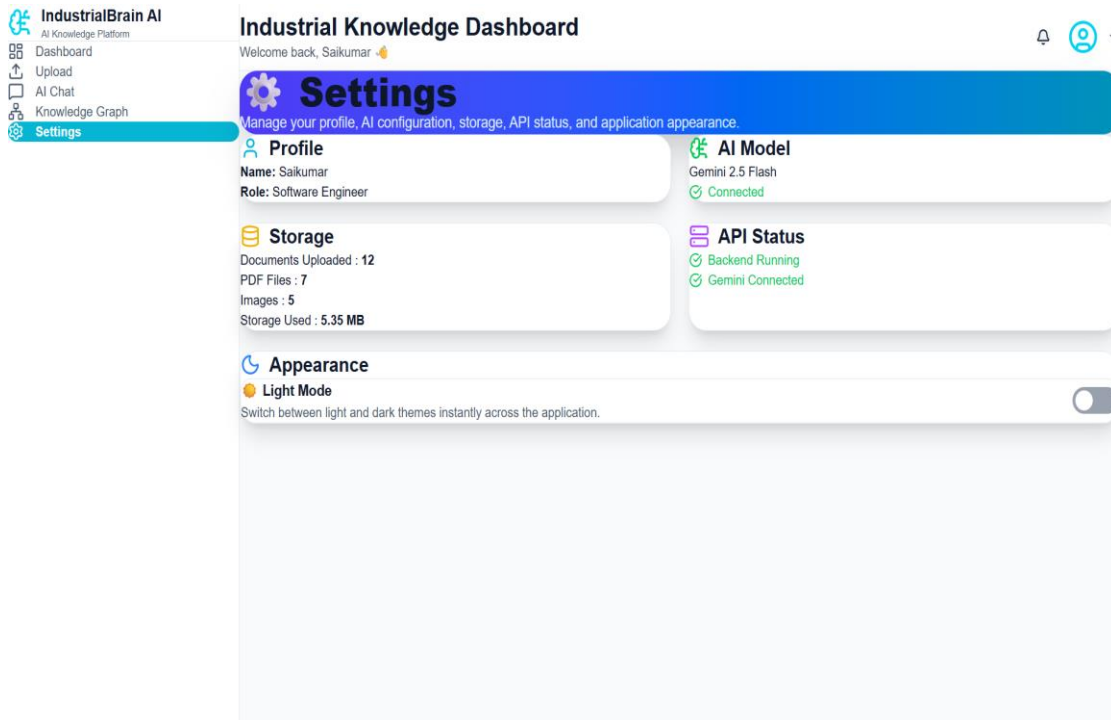


Figure 27: Light Theme

13. Security Features

JWT-based authentication and authorization.

Protected REST API endpoints.

Password encryption using Spring Security.

Role-based access to authenticated resources.

Secure document storage using Cloudinary.

Input validation and exception handling.

HTTPS-ready architecture for secure communication.

14. Challenges Faced and Solutions

Challenge	Solution
OCR Accuracy on Scanned Documents	Image preprocessing using Tesseract OCR
Technical Document Understanding	Google Gemini AI integration
Visualizing Relationships	React Flow Knowledge Graph
Secure Document Handling	JWT Authentication + Cloudinary Storage
Scalability	Modular Client–Server Architecture

15. Future Enhancements

IndustrialBrain AI has been designed with extensibility in mind. The current implementation establishes a strong foundation for intelligent industrial knowledge management while allowing future capabilities to be integrated with minimal architectural changes.

Support multilingual OCR and AI-powered document analysis.

Enable voice-based interaction with industrial documents.

Provide real-time collaboration for multiple users.

Integrate predictive maintenance recommendations using machine learning.

Support additional document formats including CAD drawings and spreadsheets.

Introduce advanced analytics dashboards with KPI reporting.

Implement role-based access control for enterprise deployments.

Deploy using Docker and Kubernetes for scalable cloud infrastructure.

Integrate IoT sensor data with knowledge graphs for contextual insights.

Provide mobile application support for field engineers.

16. Conclusion

IndustrialBrain AI demonstrates how Artificial Intelligence can transform traditional industrial document management into an intelligent knowledge platform. By combining OCR, Generative AI, Knowledge Graph visualization, secure cloud storage, and modern web technologies, the platform enables users to discover, understand, and utilize industrial knowledge more effectively.

The modular architecture, secure authentication, responsive interface, and scalable backend make the solution suitable for future enterprise adoption. The project successfully addresses the hackathon objective of applying AI to solve real-world industrial challenges while maintaining usability, performance, and extensibility.

IndustrialBrain AI represents a practical step toward digital transformation by reducing manual effort, improving information accessibility, and enabling faster decision-making through AI-driven insights.

17. References

Official Google Gemini AI Documentation

Official Tesseract OCR Documentation

Official React Documentation

Official Spring Boot Documentation

Official Spring Security Documentation

Official React Flow Documentation

Official Tailwind CSS Documentation

Official PostgreSQL Documentation

Official Cloudinary Documentation

Official JWT (JSON Web Token) Documentation

Official Git Documentation

Official GitHub Documentation

18. Repository Structure

IndustrialBrain-AI

```
|  
├─ frontend  
├─ backend  
├─ architecture  
├─ screenshots  
├─ docs  
└─ README.md
```

19. Acknowledgements

The successful development of IndustrialBrain AI was made possible through the support of modern open-source technologies, official documentation, and the Economic Times AI Hackathon 2.0 platform, which encouraged innovation in solving real-world industrial challenges.

Prepared By

Saikumar Kadiri

Project

IndustrialBrain AI

Submission

Economic Times AI Hackathon 2.0